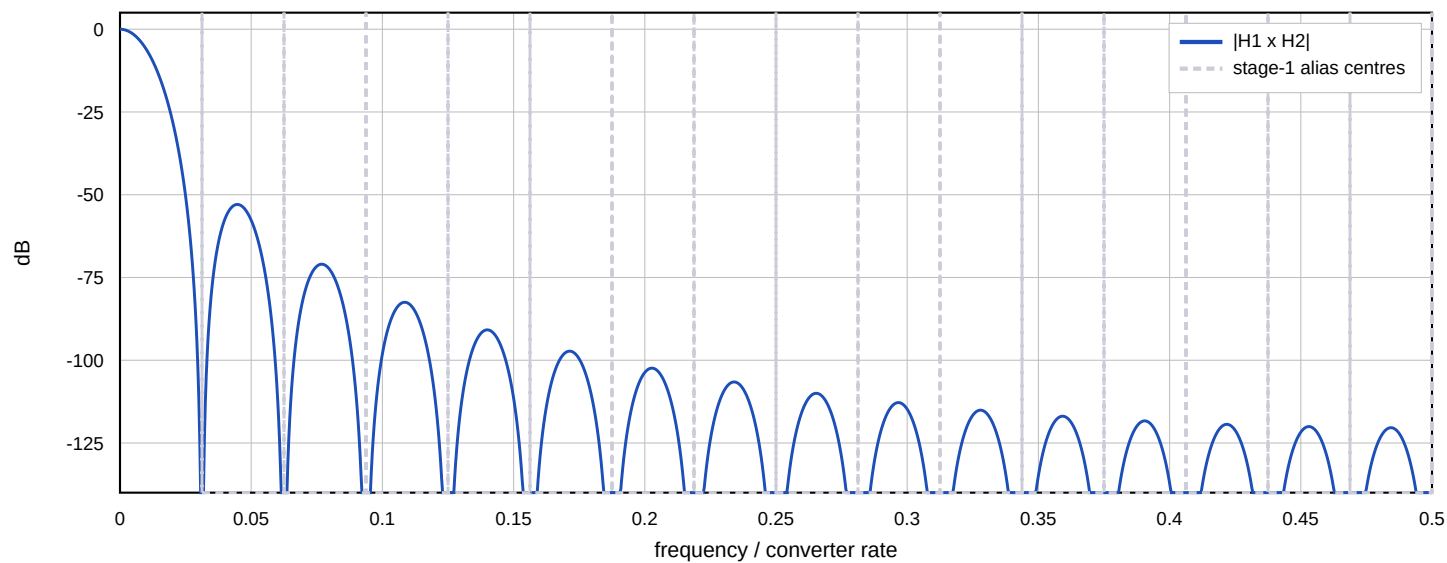


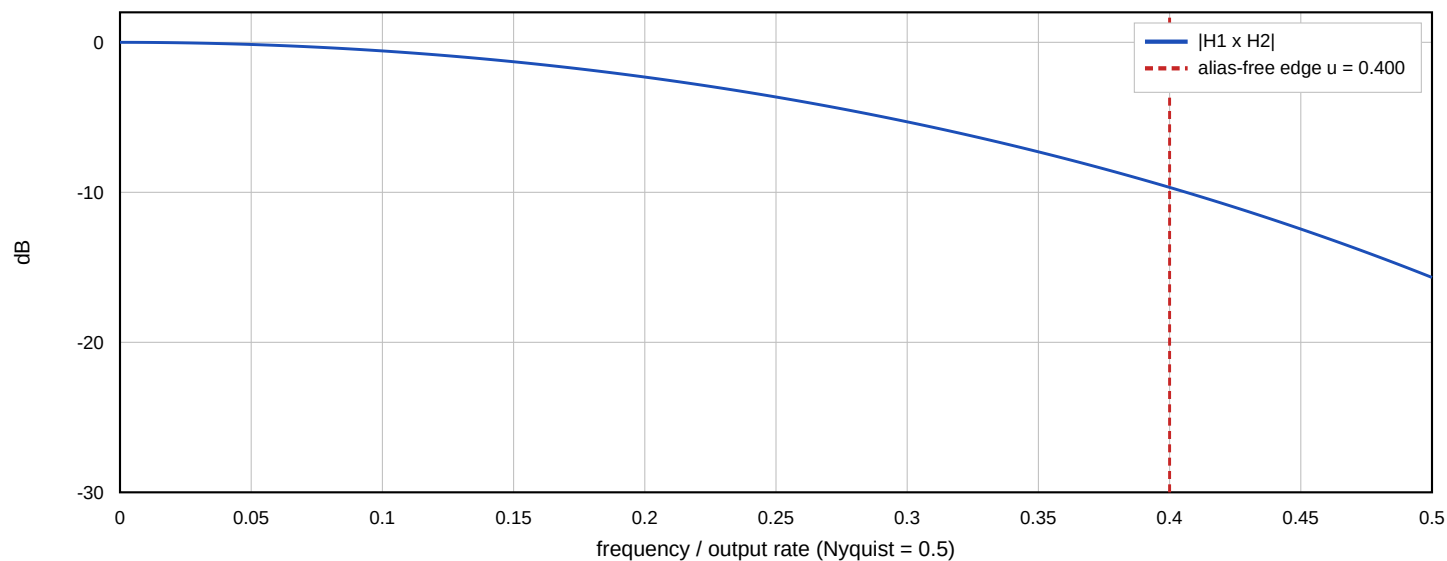
Decimation Chain - Specified Parameters

125.000 MHz / 32 = 3906.2500 kHz, split [32], 1562.500 kHz alias-free

Combined magnitude, full input band



Combined magnitude across the decimated band



Stages

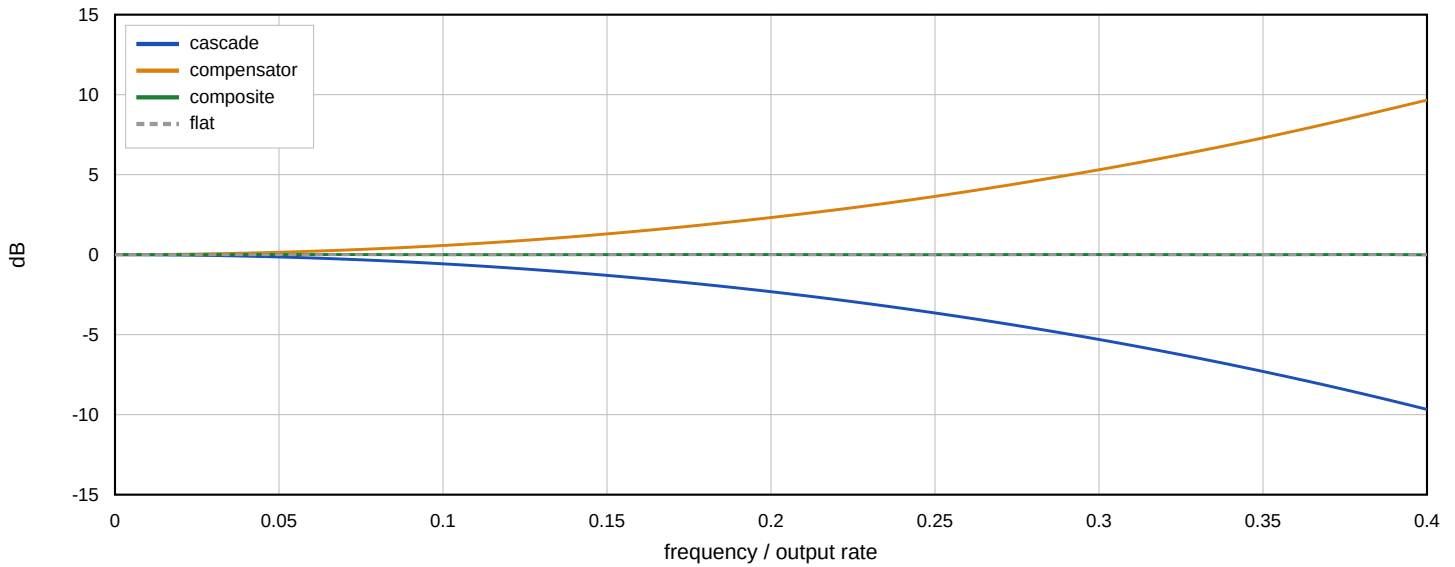
stage 1: /32 N=4 M=1 at 125.000 MHz
accumulator 36 bits, prune budget 20, widths [35, 30, 26, 23, 21, 20, 19, 18]
192 register bits, in 16 bits out 18 bits

A stage's nulls must reject the aliases of its own decimation: once energy has folded into the band no later stage can remove it, so every stage carries the full requirement.

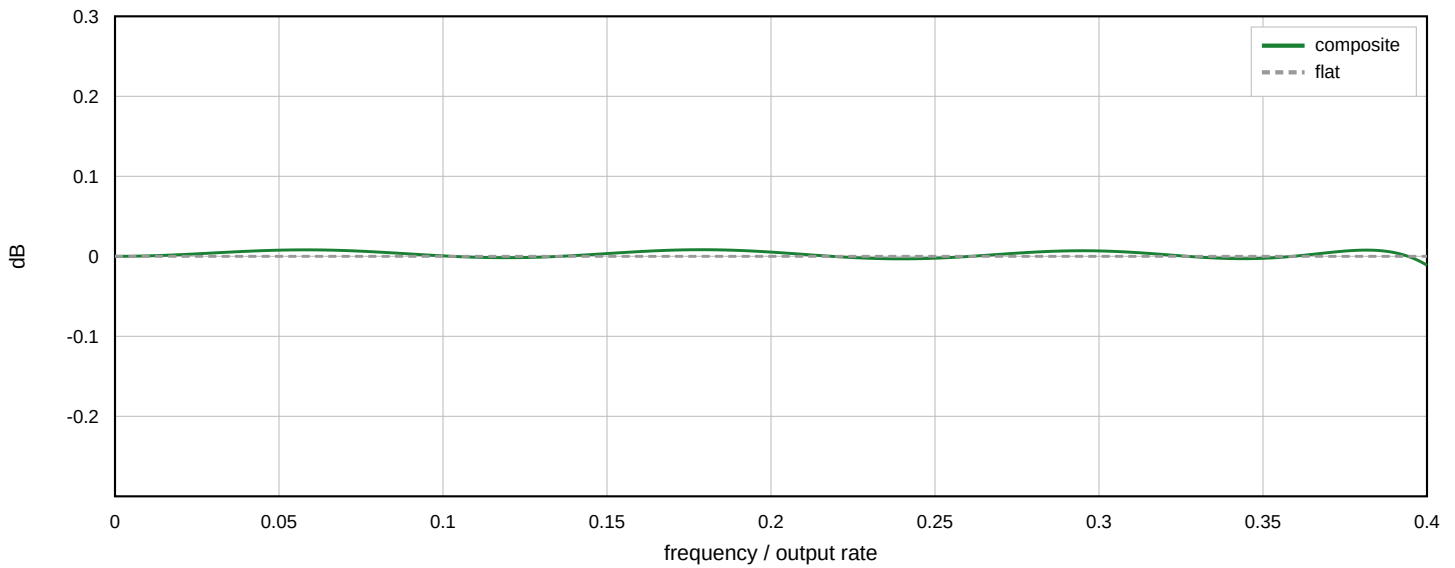
Compensation and Result

15 taps, 16-bit coefficients, 8 multipliers

Cascade, compensator, and the two together



Composite, zoomed - what is left



Achieved

ripple 0.0195 dB (asked ≤ 0.019)

alias rejection ... 23.7 dB (asked ≥ 23.7)

output SNR 99.0 dB (asked ≥ 99.0)

group delay 386 input samples = 3.1 μ s

cascade 62, integrator pipeline 3, comb pipeline 96, output regs 1, compensator 224

largest term is the compensator at 224 (58%); loop bandwidth ~ 32.4 kHz

register bits 192 (rate-weighted cost 192.0)

taps [-37, 138, -373, 887, -1998, 4545, -11226, 32512, -11226, 4545, -1998, 887, -373, 138, -37]

DC gain 1.000000 (exact by construction)

alternative none considered

Rate-weighted cost is a proxy, not an area figure: flops cost the same however slowly they are clocked. It captures that a wide adder at the converter rate is the hard part.